

Understanding ACF and PACF: A Step-by-Step Example

In time series analysis, the **Autocorrelation Function (ACF)** and the **Partial Autocorrelation Function (PACF)** are essential tools for identifying dependencies in data. To understand the difference between ACF and PACF, we consider a realistic example.

Sample Data: Daily Ice Cream Sales

Suppose a shop records the following daily sales over 10 days:

Day	Sales
1	50
2	55
3	60
4	58
5	65
6	62
7	70
8	68
9	72
10	75

Step 1: Compute the Mean

The mean of the series is calculated as:

$$\bar{X} = \frac{50 + 55 + 60 + 58 + 65 + 62 + 70 + 68 + 72 + 75}{10} = 63.5$$

Step 2: Calculating ACF

The ACF for lag k is given by:

$$\text{ACF}(k) = \frac{\sum_{t=k+1}^n (X_t - \bar{X})(X_{t-k} - \bar{X})}{\sum_{t=1}^n (X_t - \bar{X})^2}$$

Lag 1 calculation:

Sum of products = 314.0

Denominator: $\sum_{t=1}^{10} (X_t - \bar{X})^2 = 450.5$

$$\text{ACF}(1) = \frac{314}{450.5} \approx 0.696$$

t	X_t	X_{t-1}	$X_t - \bar{X}$	$X_{t-1} - \bar{X}$	Product
2	55	50	-8.5	-13.5	114.75
3	60	55	-3.5	-8.5	29.75
4	58	60	-5.5	-3.5	19.25
5	65	58	1.5	-5.5	-8.25
6	62	65	-1.5	1.5	-2.25
7	70	62	6.5	-1.5	-9.75
8	68	70	4.5	6.5	29.25
9	72	68	8.5	4.5	38.25
10	75	72	11.5	8.5	97.75

t	X_t	X_{t-2}	$X_t - \bar{X}$	$X_{t-2} - \bar{X}$	Product
3	60	50	-3.5	-13.5	47.25
4	58	55	-5.5	-8.5	46.75
5	65	60	1.5	-3.5	-5.25
6	62	58	-1.5	-5.5	8.25
7	70	65	6.5	1.5	9.75
8	68	62	4.5	-1.5	-6.75
9	72	70	8.5	6.5	55.25
10	75	68	11.5	4.5	51.75

Lag 2 ACF

$$\text{ACF}(2) = \frac{206.0}{450.5} \approx 0.458$$

Step 3: PACF Calculation

The PACF measures the **direct correlation** between X_t and X_{t-k} , after removing the effects of intermediate lags. For example:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t$$

Here, ϕ_2 represents the PACF at lag 2. PACF can be computed using regression methods.

Step 4: All ACF and PACF

The results for ACF and PACF can be summarized as:

Lag	ACF	PACF
1	0.696	0.696
2	0.458	0.236
3	0.321	0.045
4	0.210	-0.025